

Master Thesis
BlueRov2 Formation Control

Week 5

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Next Steps

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents

- BlueRov2 model no input constraints
  -
- [Research challenge] integrate input constraints
  -
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)

Achievements

- Study
  - Graph Theoretic Methods in Multiagent Networks
  - Esteban’s thesis
  - Khalil's book on Nonlinear Control

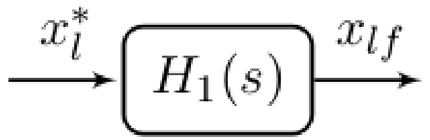
- Implementation
  - Implemented command-filtered backstepping for double-integrator agents

Command-filtered backstepping

Control input

$$u = -c_\varrho \tilde{x}_\varrho + \omega_n \alpha_{\varrho-2,2} - \tilde{x}_{\varrho-1}.$$

Command filter



$$H_1(s) := \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$

Virtual control

No constraints

$$-c_1 \sum_{j \in \mathcal{N}_i} [(x_i - x_j) - (o_i - o_j)]$$

Connectivity maintenance

$$-c_1 \sum_{j \in \mathcal{N}_i} \left[(x_i - x_j) - (o_i - o_j) + \left(\frac{1}{\Delta_k^2 - |z_k|^2} \right) z_k \right]$$

Connectivity maintenance & Collision Avoidance

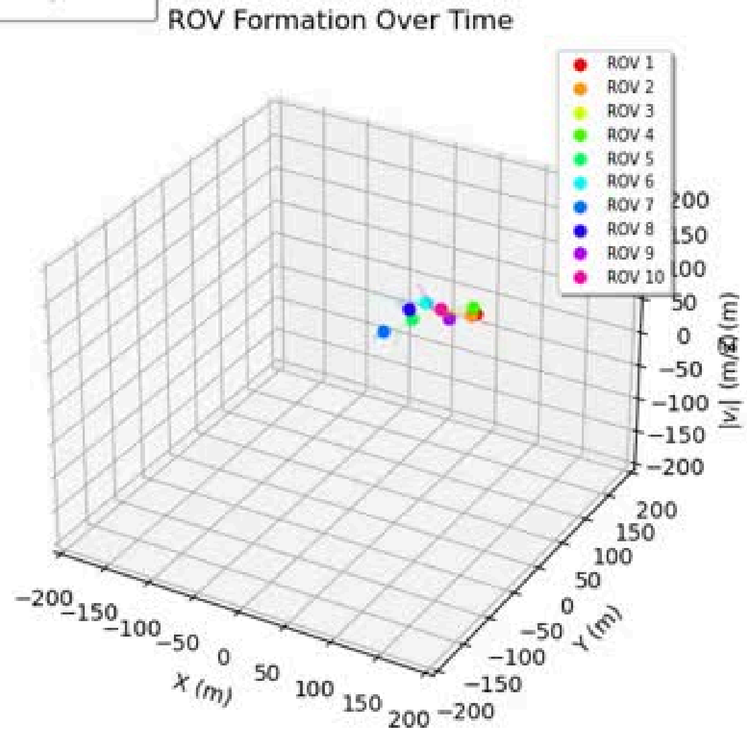
$$-c_1 \sum_{j \in \mathcal{N}_i} \left[(x_i - x_j) - (o_i - o_j) + \left[\left(\frac{1}{\Delta_k^2 - |z_k|^2} \right) - \left(\frac{1}{|z_k|^2 - \delta_k^2} \right) \right] z_k \right]$$

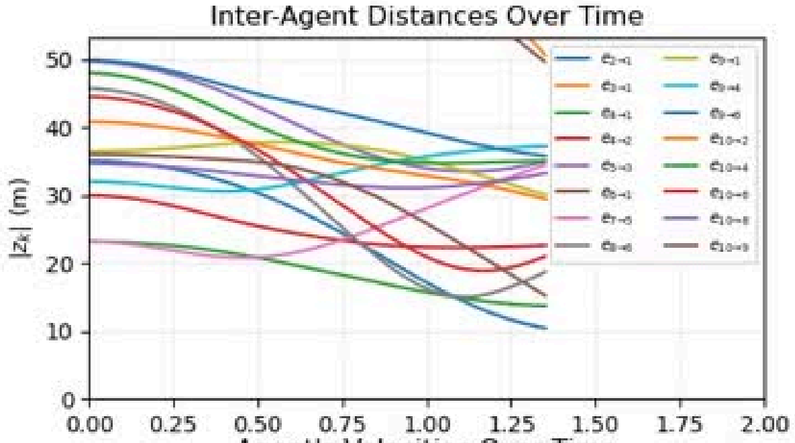
$\tilde{x}_{\varrho-1}$

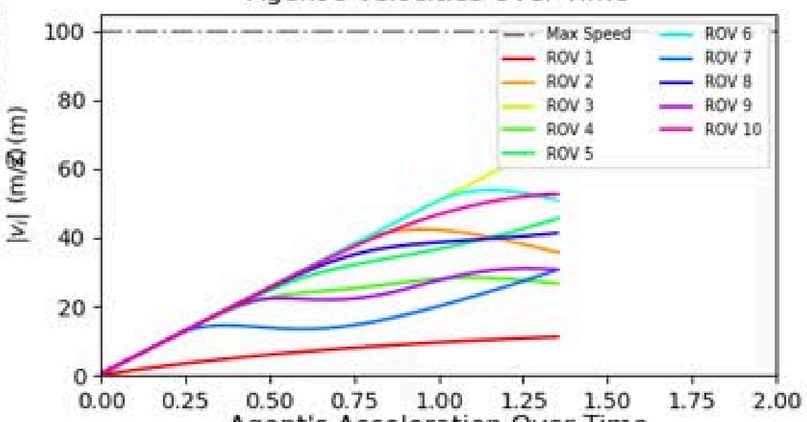
Filtered velocity

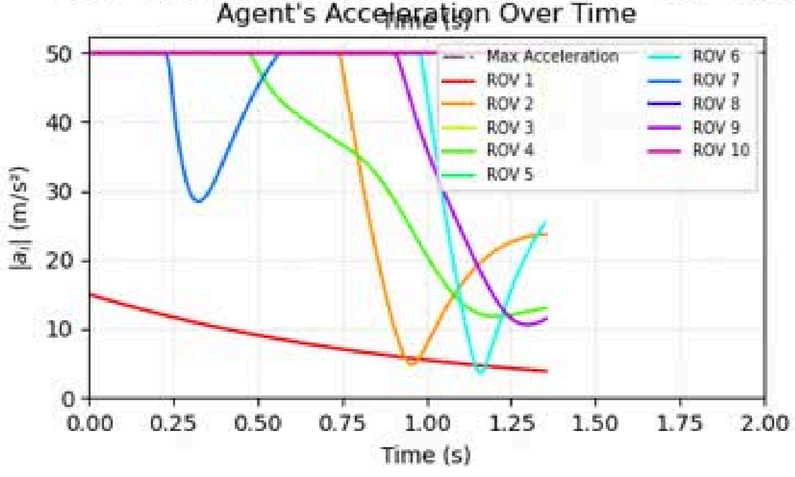
$$\tilde{x}_\varrho = v_i - x_{lf}$$

```
Time:  1.35s
----------------
ROV 1 Vel: 11.21 m/s
ROV 2 Vel: 35.81 m/s
Distance to Leader: 10.51m (Target: 20.00m)
ROV 3 Vel: 68.25 m/s
Distance to Leader: 50.73m (Target: 20.00m)
ROV 4 Vel: 26.66 m/s
Distance to Leader: 13.77m (Target: 20.00m)
ROV 5 Vel: 45.53 m/s
Distance to Leader: 72.89m (Target: 20.00m)
ROV 6 Vel: 50.83 m/s
Distance to Leader: 49.73m (Target: 20.00m)
ROV 7 Vel: 30.61 m/s
Distance to Leader: 106.09m (Target: 20.00m)
ROV 8 Vel: 41.33 m/s
Distance to Leader: 64.41m (Target: 20.00m)
ROV 9 Vel: 30.73 m/s
Distance to Leader: 30.17m (Target: 20.00m)
ROV 10 Vel: 52.61 m/s
Distance to Leader: 33.78m (Target: 20.00m)
```

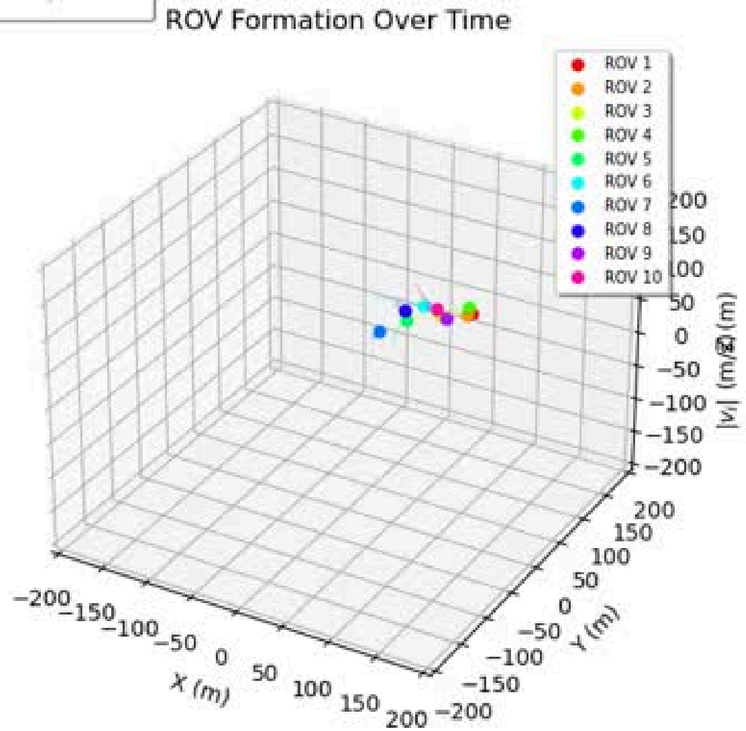


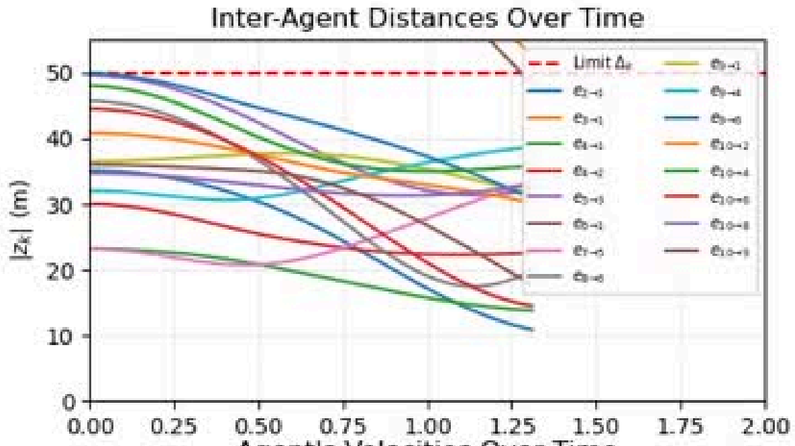


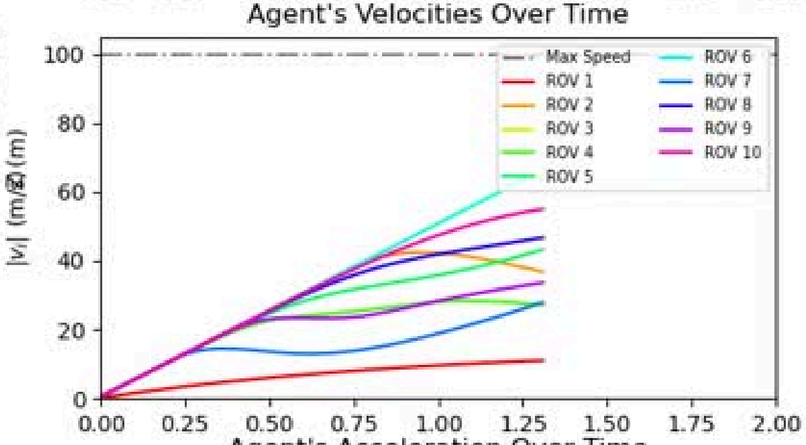


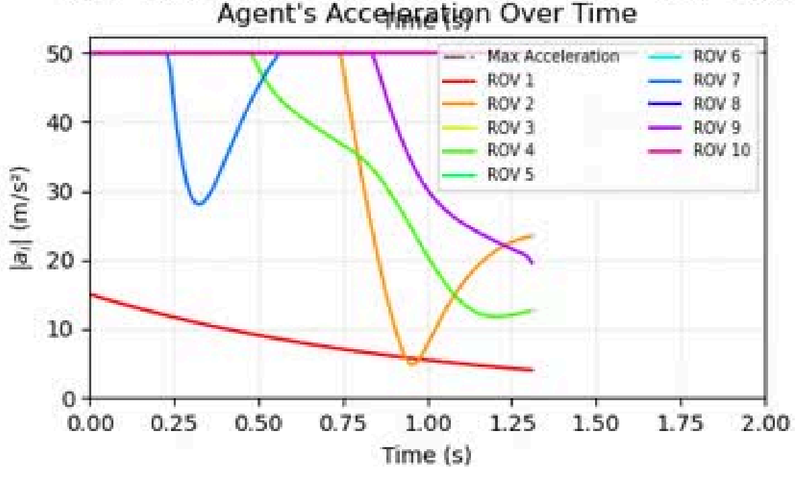


```
Time:  1.31s
---------------
ROV 1 Vel: 11.06 m/s
ROV 2 Vel: 36.73 m/s
Distance to Leader: 10.91m (Target: 20.00m)
ROV 3 Vel: 66.34 m/s
Distance to Leader: 52.45m (Target: 20.00m)
ROV 4 Vel: 27.08 m/s
Distance to Leader: 13.85m (Target: 20.00m)
ROV 5 Vel: 43.28 m/s
Distance to Leader: 74.68m (Target: 20.00m)
ROV 6 Vel: 63.40 m/s
Distance to Leader: 48.28m (Target: 20.00m)
ROV 7 Vel: 27.89 m/s
Distance to Leader: 106.91m (Target: 20.00m)
ROV 8 Vel: 46.70 m/s
Distance to Leader: 65.92m (Target: 20.00m)
ROV 9 Vel: 33.64 m/s
Distance to Leader: 31.56m (Target: 20.00m)
ROV 10 Vel: 54.96 m/s
Distance to Leader: 35.38m (Target: 20.00m)
```

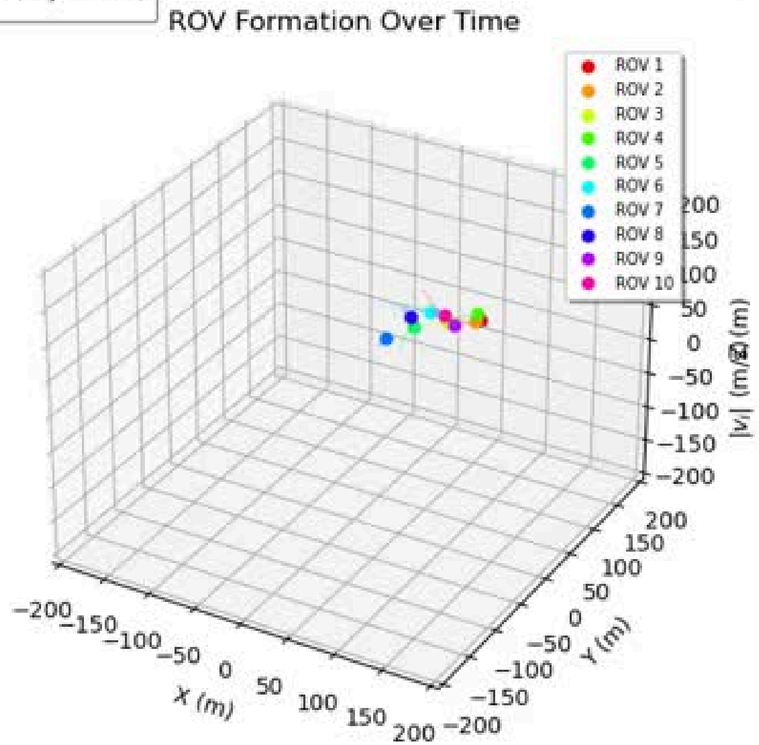


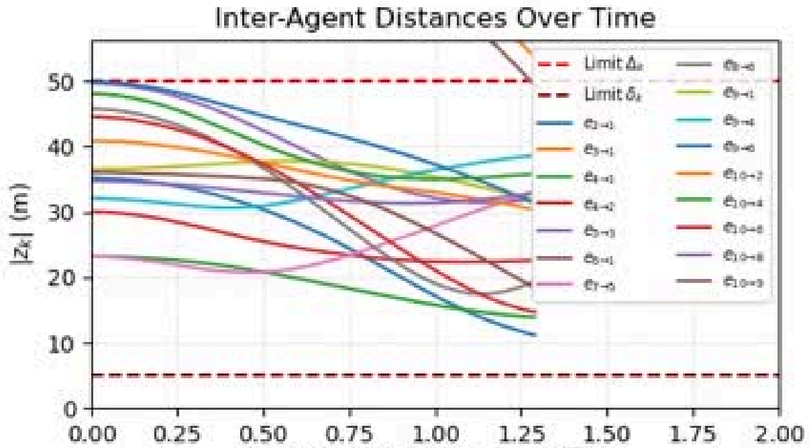


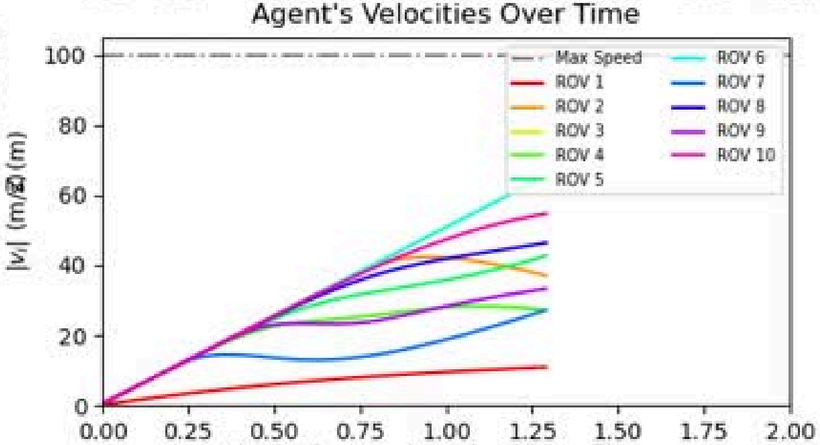


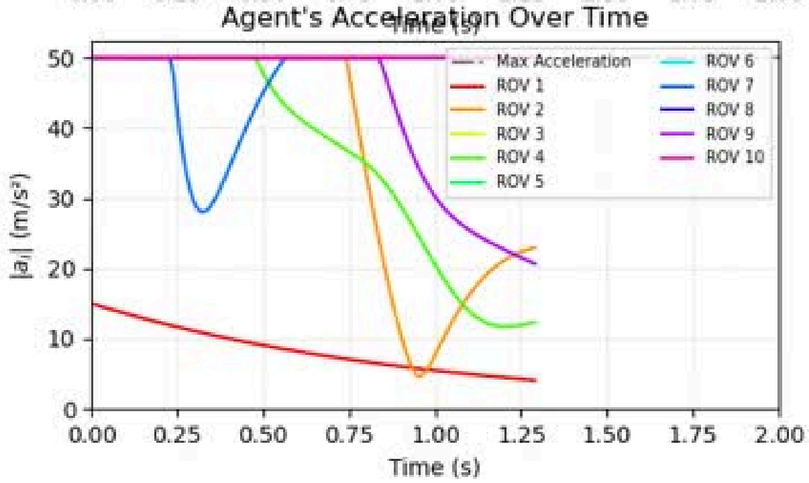


```
Time:  1.29s
----------------
ROV 1 Vel: 10.98 m/s
ROV 2 Vel: 37.11 m/s
Distance to Leader: 11.20m (Target: 20.00m)
ROV 3 Vel: 65.36 m/s
Distance to Leader: 53.58m (Target: 20.00m)
ROV 4 Vel: 27.25 m/s
Distance to Leader: 13.96m (Target: 20.00m)
ROV 5 Vel: 42.78 m/s
Distance to Leader: 75.20m (Target: 20.00m)
ROV 6 Vel: 64.40 m/s
Distance to Leader: 49.35m (Target: 20.00m)
ROV 7 Vel: 27.23 m/s
Distance to Leader: 107.17m (Target: 20.00m)
ROV 8 Vel: 46.37 m/s
Distance to Leader: 66.29m (Target: 20.00m)
ROV 9 Vel: 33.35 m/s
Distance to Leader: 31.83m (Target: 20.00m)
ROV 10 Vel: 54.86 m/s
Distance to Leader: 35.94m (Target: 20.00m)
```









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  - Undirected graph
  - Directed graph
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  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents

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Questions & Comments